Endpoint variation in orthogonal symmetry markings

Human Compression project: L031 corrected recovery

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We rework the positive characteristic-two construction of Brooksbank–Ferrara–Leemans [BFL]. Full orders are imposed on the interior bonds of a path; its two endpoint coefficients may vary. This supplies both orthogonal types without an unstated choice between primitive torus parameters. Every group below acts on its actual ambient space; restriction to a degenerate subspace is not assumed faithful. The complete source contract is supplied by the companion source-interface paper and its explicitly acquired low-dimensional and binary proofs.

1 Classical inputs and support groups

Let $F = \mathbf{F}_q$, $q = 2^k \geq 4$. A quadratic space will be called *nonsingular* here when its polar form B is nondegenerate, or its radical is one-dimensional and has nonzero quadratic value. For $Q(v) \neq 0$ write

$$s_v(x) = x + \frac{B(x,v)}{Q(v)}v, \qquad \Sigma_U = \langle s_v : v \in U,\ Q(v) \neq 0 \rangle.$$

An orthogonal subspace has polar radical of dimension at most one. The following standard classical inputs remain explicit: (i) in a nondegenerate even-dimensional quadratic space over F, the full orthogonal group is generated by its symmetries; (ii) the Dickson character of this group is $g \mapsto \operatorname{rank}(g-1) \pmod 2$; (iii) the nonsplit dihedral subgroup of order $2(q+1)$ is maximal in $\mathrm{SL}_2(q)$ for $q \geq 4$; and (iv) the Arf trace distinguishes the two nondegenerate even-dimensional quadratic forms. These are classical group and quadratic-form facts, not invocations of the construction theorem being recovered. Input (i) is the symmetry-generation theorem cited by [BFL, §3] from Taylor, Theorem 11.39. Input (ii) is Taylor, Theorem 11.43 (printed p.160). Input (iii) is the rank-one subgroup classification used in [BFL, Lemma 3.4] and stated in [LMO, Proposition 2.3, p.5]: a subgroup containing an element of order $q+1$ cannot lie in a Borel, split torus normalizer, or proper subfield group, and the remaining nonsplit torus normalizer is the given one. Their full proofs remain external acquisition costs.

We will use two elementary consequences. Symplectic groups over F are generated by their transvections $t_{u,a}(x) = x + aB(x,u)u$, $a \neq 0$. Indeed such maps carry any nonzero vector to any other: use the transvection in their sum if their polar product is nonzero, and an intermediate vector having nonzero polar products with both otherwise. After fixing a vector e, two possible partners f, f' with $B(e,f) = B(e,f') = 1$ are joined by transvections with axes in $e^{\perp}$. If necessary use $f'' = f' + te$ where $t \notin \{0, B(f,f')\}$; the two polar products are then nonzero. Having fixed a hyperbolic pair, induct on its perpendicular complement. Also $\mathrm{SL}_2(q)$ is perfect: its upper and lower elementary unipotents generate it, and conjugation by $\operatorname{diag}(a, a^{-1})$, $a^2 \neq 1$, expresses every such unipotent as a commutator. Both assertions require no finite census.

Lemma 1 (Support characterization)**.** *Let V be nonsingular as defined above. For an orthogonal subspace $U \leq V$ of dimension at least two put*

$$K(U) = \{g \in O(V) : \operatorname{im}(g-1) \subseteq U\}.$$

Then $K(U) = \Sigma_U$. For a nonsingular one-space $U = \langle v \rangle$ the same equality is $K(U) = \langle s_v \rangle$. Consequently if $U, W, U \cap W$ are orthogonal subspaces of dimension at least two, then $\Sigma_U \cap \Sigma_W = \Sigma_{U \cap W}$. The conclusion also holds when $U \cap W$ is a nonsingular one-space.

Proof. Every product of symmetries in U has support in U. Conversely, an isometry with support in U fixes $U^\perp$ pointwise. To see this, for $y \in U^\perp$, $B(x, (g^{-1} - 1)y) = B((g-1)x, y) = 0$ for every x. Thus $(g^{-1} - 1)y$ is in the polar radical of V. It is zero when the radical is zero; otherwise preservation of Q and the nonzero quadratic value of its generator force it to be zero. In particular $K(U)$ preserves U and fixes its radical.

If U is nondegenerate, $V = U \perp U^\perp$ and $K(U)$ is its full orthogonal group extended trivially; input (i) applies. Suppose next that $U = M \perp \langle z \rangle$, with M nondegenerate and $Q(z) = 0$. This z is not radical in V. Choose $w \in M^\perp$ with $B(z, w) = 1$, and replace w by $w + Q(w)z$ to make it singular. On $M \perp \langle z, w \rangle$, the kernel of the induced map $K(U) \longrightarrow O(M)$ consists exactly of the maps

$$T_u(x) = x + B(x, z)u + \big(B(x, u) + Q(u)B(x, z)\big)z, \qquad u \in M. \tag{1}$$

On its perpendicular complement these maps are the identity. Indeed the action on M must be $x \mapsto x + B(x, u)z$; preserving the pairing with w determines its M-component u, and preserving $Q(w)$ determines the remaining coefficient $Q(u)$. Direct substitution gives $T_u T_v = T_{u+v}$. For $v \in M$ nonsingular and $a \in F$ one has

$$s_v s_{v+az} = T_{av/Q(v)}. \tag{2}$$

Thus every scalar multiple of every nonsingular v occurs as a translation parameter. Such vectors span M: in a hyperbolic plane one may replace its singular basis by $e + f, e + af$ with $a \notin \{0, 1\}$, and an asingular plane already has a nonsingular basis. Input (i), equations (1)–(2), and the trivial extensions of $O(M)$ now give $K(U) = \Sigma_U$.

If instead $Q(z) \neq 0$, the induced map $K(U) \longrightarrow \mathrm{Sp}(U/\langle z \rangle)$ is onto already on Σ_U. In $U = M \perp \langle z \rangle$, the transvection with axis $u \in M$ and coefficient $a \neq 0$ is induced by s_{u+tz}, where $Q(u) + t^2 Q(z) = a^{-1}$; the square map on F is bijective. Use the elementary transvection generation proved above. An element of the kernel fixes U pointwise, since a nonzero radical shear inside U would change its quadratic value. If z is radical in V, this kernel is trivial. Otherwise an element fixing U pointwise and having support in U has the form $x \mapsto x + cB(x, z)z$. Preservation of Q gives $c^2 Q(z) + c = 0$, so the kernel is exactly $\{1, s_z\}$, also contained in Σ_U.

For a nonsingular one-space, write a supported map as $g(x) = x + f(x)v$. Quadratic preservation says $f(x)(Q(v)f(x) + B(x, v)) = 0$ for every x. A vector space is not the union of two proper linear subspaces, so either $f = 0$ or $f = B(-, v)/Q(v)$. Finally intersection of two support conditions is precisely support in the intersection. This proves every asserted equality. □

Lemma 2 (An ambient three-space)**.** *Let $T \leq V$ be an orthogonal three-space, let $L \leq T$ be asingular of dimension two, and let $x \in T \setminus L$ be nonsingular and nonradical in T. Then*

$$\langle \Sigma_L, s_x \rangle = \Sigma_T.$$

Proof. Write $T = L \perp \langle z \rangle$. If $Q(z) = 0$, write $x = u + tz$ with $u \neq 0$, $t \neq 0$. Equation (2) supplies a nonzero translation in $K(T) = L^+ \rtimes O^-(L)$. The cyclic torus of order $q + 1$ acts irreducibly on L even as an $\mathbf{F}_2$-space: an element of that order in $\mathbf{F}_{q^2}$ has degree $2k$ over $\mathbf{F}_2$, since a proper divisor of $2k$ is at most k and $q + 1 > 2^k - 1$. Its orbit therefore spans every translation parameter over $\mathbf{F}_2$. This gives all of $K(T)$.

Suppose $Q(z) \neq 0$. The projection of Σ_L to $\mathrm{Sp}(T/\langle z \rangle) = \mathrm{SL}_2(q)$ is its nonsplit dihedral group. The projection of s_x, for $x = u + tz$, has axis u and transvection coefficient $(Q(u) + t^2 Q(z))^{-1}$. This differs from the coefficient $Q(u)^{-1}$ of the unique dihedral involution on that axis. Hence it is outside that dihedral group, and input (iii) gives the full $\mathrm{SL}_2(q)$ image.

If z is radical in V, projection is injective and we are done. Otherwise the kernel is $\langle s_z \rangle$ by Lemma 1. The Dickson character on the nondegenerate even space obtained by adjoining a partner to z restricts to $K(T)$ and takes value one on s_z. Thus $K(T)$ is the direct product of its character kernel, isomorphic to $\mathrm{SL}_2(q)$, and $\langle s_z \rangle$. A subgroup projecting bijectively to this perfect group would have trivial Dickson character. Our subgroup contains symmetries, of character one, so its projection kernel is nontrivial and it equals $K(T) = \Sigma_T$. □

Lemma 3 (Hyperplane saturation)**.** *Let W be an orthogonal hyperplane of a nonsingular quadratic space V, with $\dim V \geq 4$. Let L be an asingular two-space not contained in W. Then $\langle \Sigma_W, \Sigma_L \rangle = \Sigma_V$.*

Proof. For a nonsingular y outside $W \cup L$, put $T = \langle L, y \rangle$. This is an orthogonal three-space. Its intersection $M = T \cap W$ is a two-space meeting L in a nonsingular point. Among the $q + 1 \geq 5$ projective points of M, at most two are singular, at most one lies in L, and at most one is radical in T. A remaining point supplies a nonsingular nonradical $x \in M \setminus L$. Lemma 2 now gives Σ_T, including s_y. All nonsingular roots in $W \cup L$ were already available. □

2 Paths with free endpoints

For $d \geq 3$ set

$$Q(x) = \sum_{i=1}^{d} x_i^2 + \sum_{i=1}^{d-1} \alpha_i x_i x_{i+1}, \qquad \alpha_i \neq 0. \tag{3}$$

Every interval of the basis is orthogonal: its tridiagonal polar matrix is nondegenerate in even dimension and has radical dimension one in odd dimension. This follows successively from its first row; equivalently the even determinant is the square of the product of its alternate nonzero adjacent coefficients.

Choose $\lambda \neq 0$ such that

$$H_\lambda = \begin{pmatrix} 1 & \lambda \\ \lambda & 1 + \lambda^2 \end{pmatrix} \quad \text{has order } q + 1. \tag{4}$$

Such a choice always exists: take ζ of exact order $q + 1$ in the norm-one torus of $\mathbf{F}_{q^2}^\times$ and set $\lambda^2 = \zeta + \zeta^{-1} \in F$. The eigenvalues are ζ, ζ^{-1}, and are distinct. The associated two-space is asingular, since a hyperbolic plane's rotation group has order $q - 1$.

Theorem 4. *Suppose (3) is nonsingular. If $d = 3$, require at least one of its two bonds to have the full order $q + 1$ of (4). If $d \geq 4$, require every interior bond $\alpha_2, \ldots, \alpha_{d-2}$ to have that order; the two endpoint coefficients remain arbitrary nonzero elements. Then its d basis symmetries generate the full isometry group and form a string C-group of rank d.*

Proof. First prove generation of Σ_U by the marked symmetries of every interval U of length at least three. In length three use its full asingular two-space and Lemma 2; the third basis vector is nonradical because its pairing with the middle vector is nonzero.

In length $b \geq 4$, relabel the interval $v_1, \ldots, v_b$ and put $W = \langle v_2, \ldots, v_b \rangle$. By induction its marked group is Σ_W. The full asingular line $L_0 = \langle v_2, v_3 \rangle$ is available. Its conjugate $L = L_0^{s_{v_1}}$ is asingular and not contained in W, since $s_{v_1}(v_2) = v_2 + \alpha_1 v_1$. The marked group contains Σ_L by conjugation. The argument of Lemma 3, applied inside U with all symmetries still acting on the original ambient V, now gives Σ_U. It uses only orthogonality of its three-spaces, so also applies when the radical of this particular interval is singular. With $U = V$, Lemma 1 gives the full isometry group.

For intersection, every two-generator submarking is a dihedral C-group: its roots are independent, the symmetries are distinct, and each has order two. In rank three, the intersection

of the end parabolics is supported on the middle one-space and hence is exactly its symmetry subgroup by Lemma 1. For a longer interval, its two end parabolics are the full symmetry groups of their spans. Their intersection is the symmetry group of the middle span, again by Lemma 1. In rank four this middle span is the full-order interior two-space; in larger ranks its generation was just proved. Apply the standard string C-group end-parabolic criterion inductively. This criterion is the retained McMullen–Schulte Proposition 2E16, also used in the accepted L037 common contraction paper. Nonadjacent symmetries commute; adjacent ones do not because their pairing is nonzero. All string and full intersection requirements follow. □

3 Both types and the odd-dimensional group

Theorem 5. *For $q \geq 4$, the groups $O^+(2m, q)$ and $O^-(2m, q)$ have rank-$2m$ string C-markings by symmetries for every $m \geq 1$. The group $\mathrm{Sp}(2m, q)$ has one of rank $2m + 1$ for every $m \geq 1$. Every adjacent product in these markings has odd order.*

Proof. For dimension two the two orthogonal groups are the dihedral groups of orders $2(q - 1)$ and $2(q + 1)$, generated by two symmetries whose product generates the corresponding torus. Both torus orders are odd and at least three.

For even $d = 2m \geq 4$, put every interior coefficient equal to λ of (4). Set the last coefficient to $\beta \neq 0$, and consider the odd suffix $W = \langle v_2, \ldots, v_d \rangle$. Its radical generator has successive nonzero alternate coordinates

$$z = v_2 + v_4 + \cdots + v_{d-2} + (\lambda/\beta)v_d.$$

Consequently $c = Q(z) = (m - 1) + \lambda^2/\beta^2$. If m is even choose $\beta \neq \lambda$; if m is odd any nonzero β works. Thus $c \neq 0$. Fix a nondegenerate complement M to $\langle z \rangle$ in W.

Now let $\alpha = \alpha_1$ vary over $F^\times$. The pairing $B(v_1, -)$ on M is α times a fixed functional, so there is a fixed $u \in M$ for which $v_1' = v_1 + \alpha u$ is perpendicular to M. Write $Q(v_1') = 1 + \alpha^2 h$, where h is independent of α. Also $B(v_1', z) = \alpha$. The decomposition $V = M \perp \langle z, v_1' \rangle$ therefore has Arf trace

$$\mathrm{Tr}(\mathrm{Arf}\, M + ch) + \mathrm{Tr}(c/\alpha^2). \tag{5}$$

Since $c \neq 0$ and squaring is a bijection, the second term takes both values zero and one as α ranges over $F^\times$: a nonzero trace functional has $q/2$ points in each fiber, and $q/2 - 1 \geq 1$ nonzero points in its zero fiber. Both orthogonal types are thus attained. Theorem 4 applies with these free endpoint choices.

For odd $d = 2m + 1$, take $\alpha_1 = \cdots = \alpha_{d-2} = \lambda$ and $\alpha_{d-1} = \beta \neq 0$. Its radical generator is $v_1 + v_3 + \cdots + v_{d-2} + (\lambda/\beta)v_d$ and has value $m + \lambda^2/\beta^2$. Choose $\beta \neq \lambda$ when m is odd; otherwise any nonzero β works. The space is nonsingular with one-dimensional polar radical, so Theorem 4 applies (including the rank-three case). Its full isometry group is $\mathrm{Sp}(2m, q)$: the quotient map is injective because the radical is nonsingular, and the lifted transvections in Lemma 1 give surjectivity.

Finally an adjacent support is a nondegenerate two-space, and the characteristic polynomial of its product is $X^2 + \alpha_i^2 X + 1$. It has distinct roots because $\alpha_i \neq 0$, lying in F or its quadratic extension as an inverse pair. Their orders divide $q - 1$ or $q + 1$, hence are odd. On the perpendicular complement the product is the identity, so its whole-space order is the same. □

Complete source interface. Theorem 5 gives the positive existence and exact group claims of [BFL, Theorem 1.4 and Corollary 1.5], with a different, fully specified type-selection proof. It does not assert that every choice of source coefficients has nonsingular radical or generates the entire isometry group when that radical is singular. The three binary examples and higher-dimensional binary exclusion are separate, fully acquired companion proofs with distinct reviews. The source-interface and four-matrix companions, together with the acquired small-dimension obstruction and centralizer papers, prove the complete corrected low-dimensional spectrum.

References

[1] P. A. Brooksbank, J. T. Ferrara and D. Leemans, *Orthogonal groups in characteristic 2 acting on polytopes of high rank*, arXiv:1709.02219v2, 10 March 2018, supplied twelve-page edition.

[2] D. E. Taylor, *The Geometry of the Classical Groups*, Heldermann, 1992, Theorems 11.39 and 11.43, printed pp.157 and 160. Classical symmetry generation and quadratic forms remain retained inputs with their full external proof costs.

[3] D. Leemans, J. Moerenhout and E. O'Reilly-Regueiro, *Projective linear groups as automorphism groups of chiral polytopes*, supplied twenty-five-page edition, Proposition 2.3, p.5; the underlying Dickson classification remains an external input.

[4] P. McMullen and E. Schulte, *Abstract Regular Polytopes*, Cambridge, 2002, Proposition 2E16.